

AASD 4010

Deep Learning - I

Applied AI Solutions Developer Program



Transformers



Agenda



- Transformers
- Pipelines
- Finetuning
- Finetuning Text Classification
- Finetuning Token Classification
- Finetuning QA
- Finetuning Translation
- Finetuning Summarization



Transformers

Transformers is an amazing **go-to** library for all your NLP needs



Pipelines

Pipeline in Transformers are **ready-to-use** pretrained and finetuned model ready for **inferencing**

It supports the following tasks:

- Sentiment Analysis
- Named Entity Recognition
- Question Answering
- Translation
- Summarization
- Text Generation
- Feature Extraction



Pre-Training

Pre-Training of BERT involves

- Masked Language Model (**MLM**)
- Next Sentence Prediction (**NSP**)

MLM --> Predict the masked word

NSP --> Is Sentence B follow Sentence A logically?

Fake Tasks – BERT is trained to perform these tasks to force it to develop an amazing understanding of the language. Once BERT is trained, discard the tiny piece of the model and replace it with something useful

MLM & NSP **do not** need labeled data

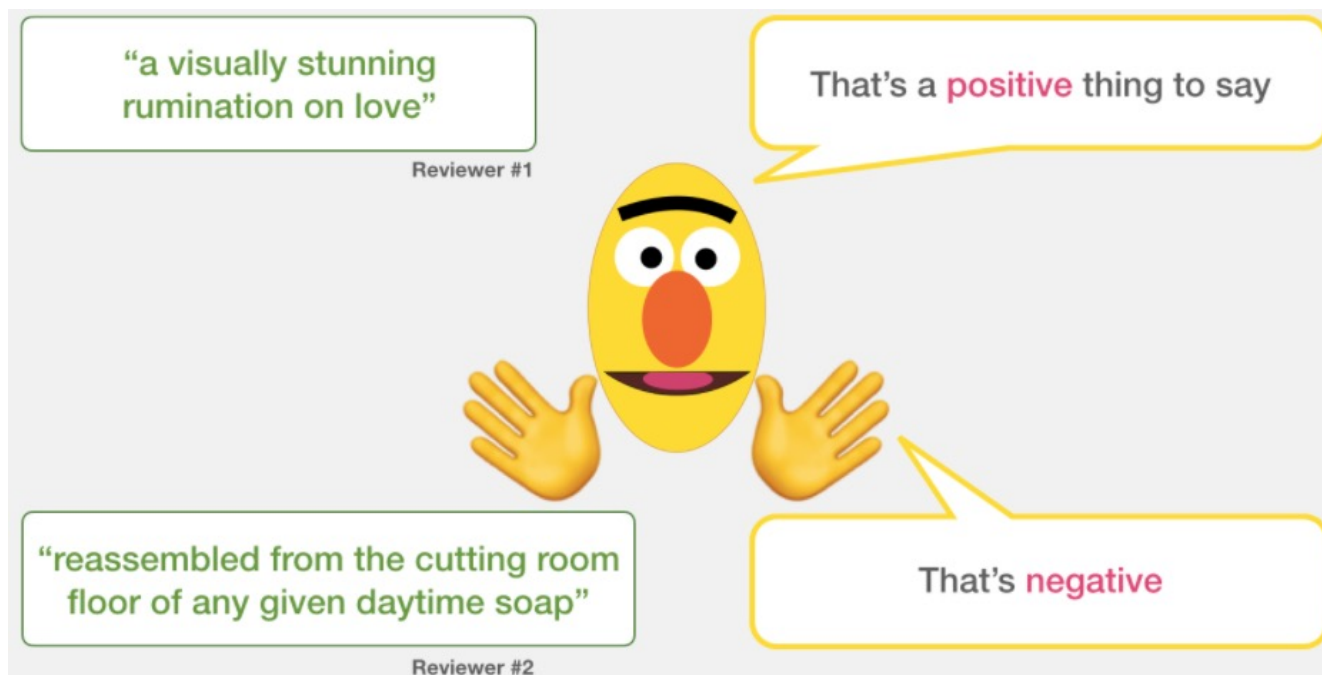


Finetuning BERT

BERT



Task: Classify positive or negative review



Reviews dataset

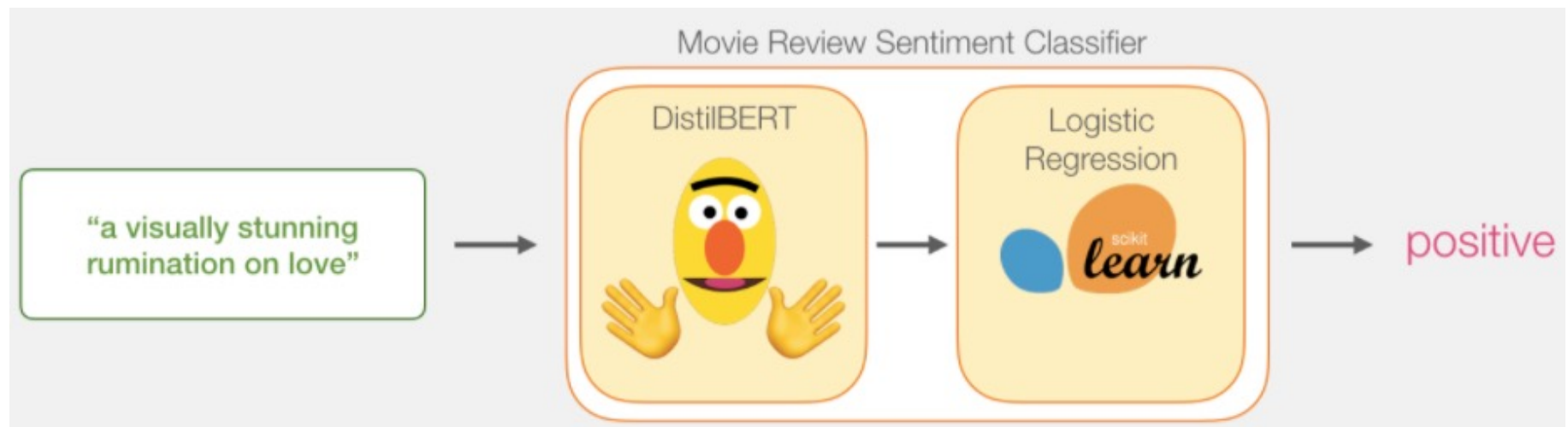
sentence	label
a stirring , funny and finally transporting re imagining of beauty and the beast and 1930s horror films	1
apparently reassembled from the cutting room floor of any given daytime soap	0
they presume their audience won't sit still for a sociology lesson	0
this is a visually stunning rumination on love , memory , history and the war between art and commerce	1
jonathan parker 's bartleby should have been the be all end all of the modern office anomie films	1



Movie Review Classifier



Pretrained model + Linear Classifier

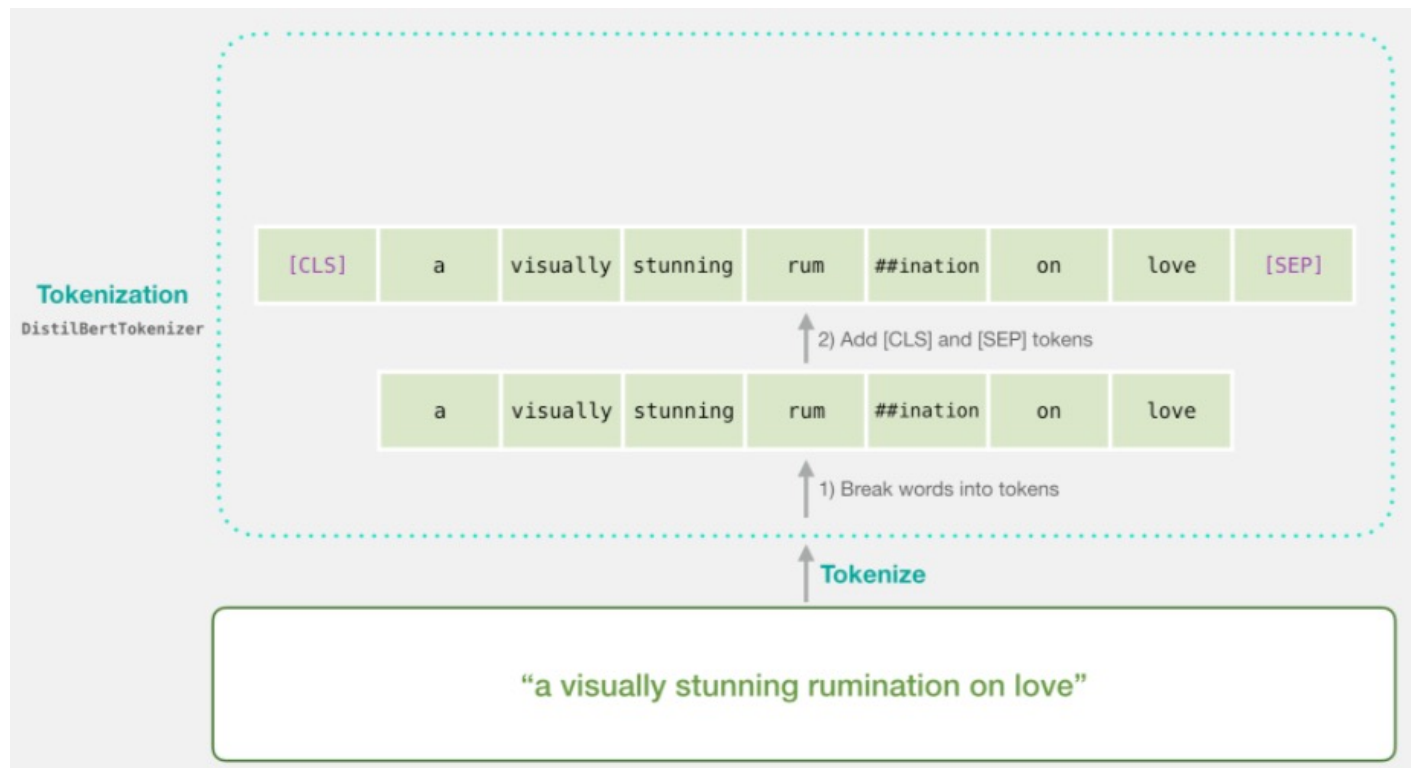


Pretrained model

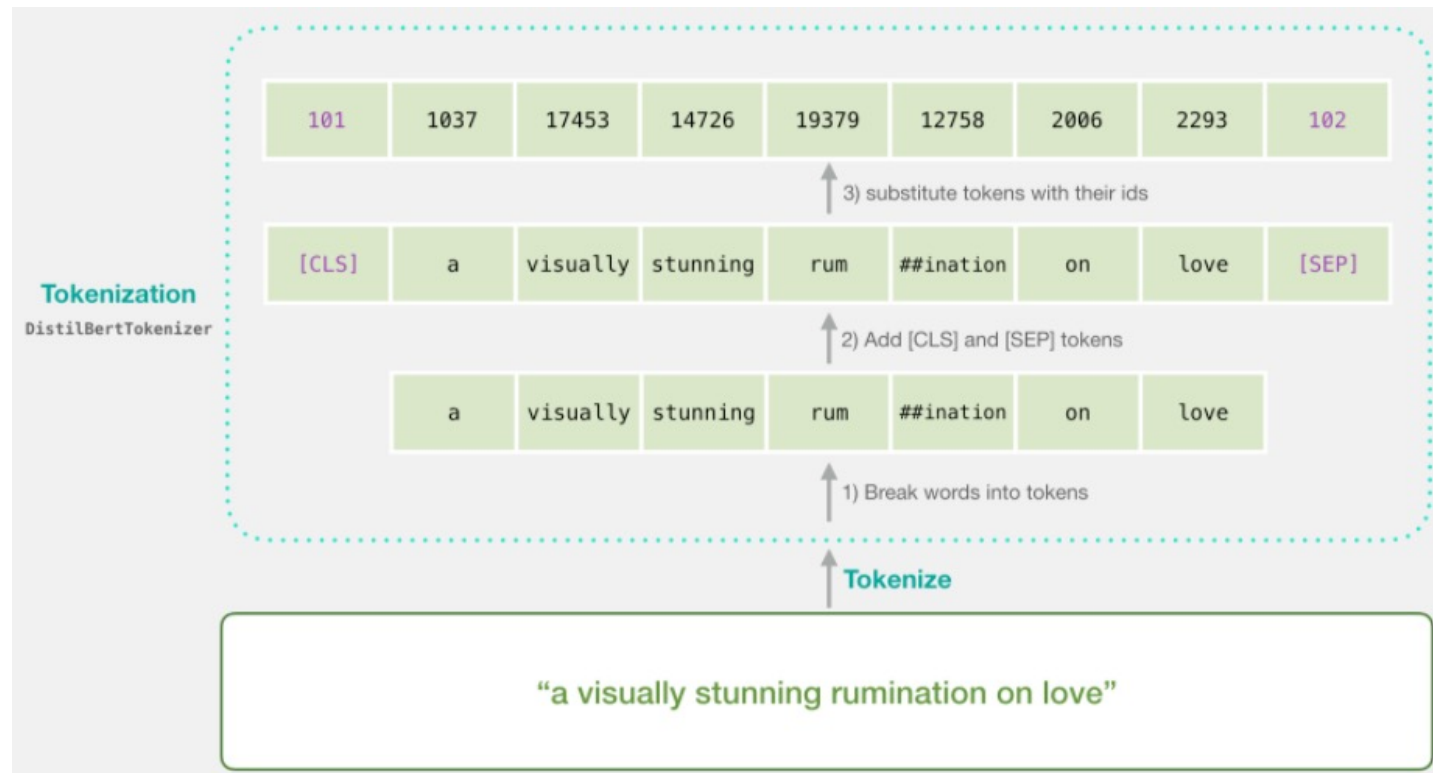
Step #1: Use DistilBERT to embed all the sentences



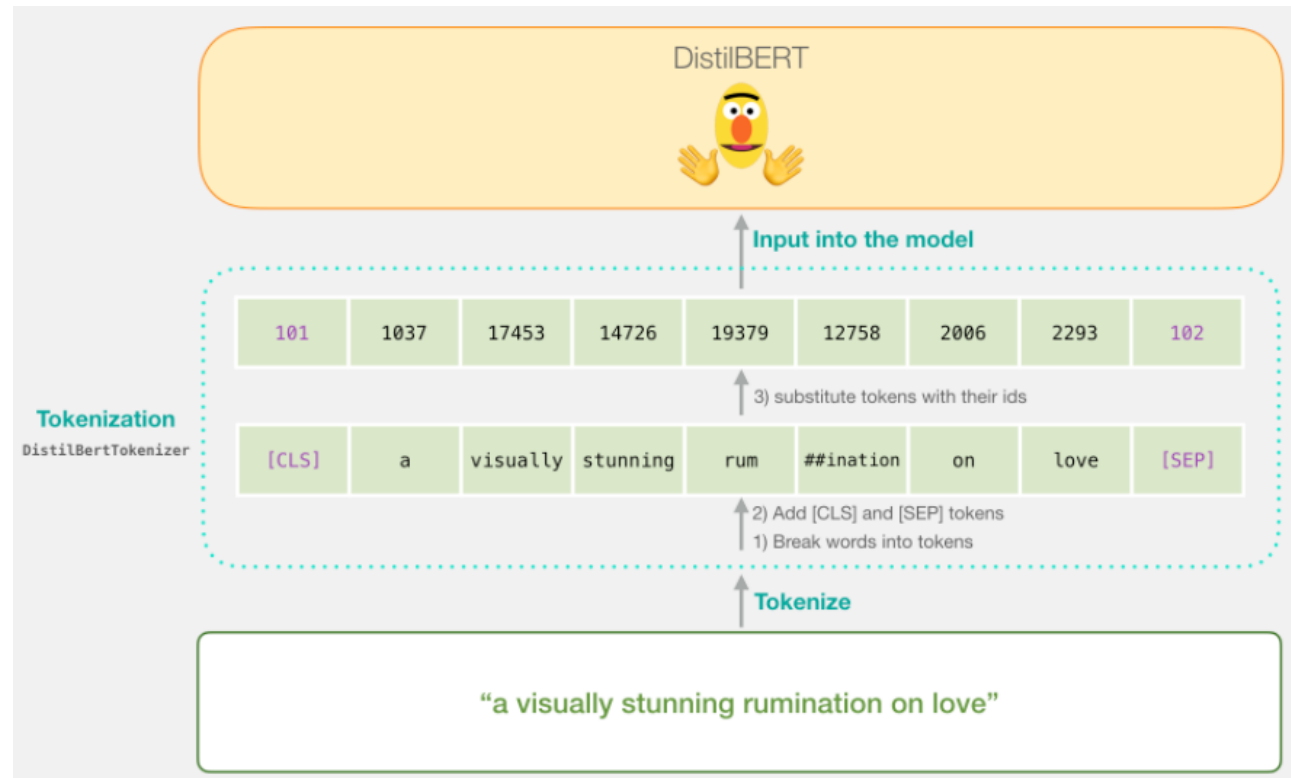
Tokenization



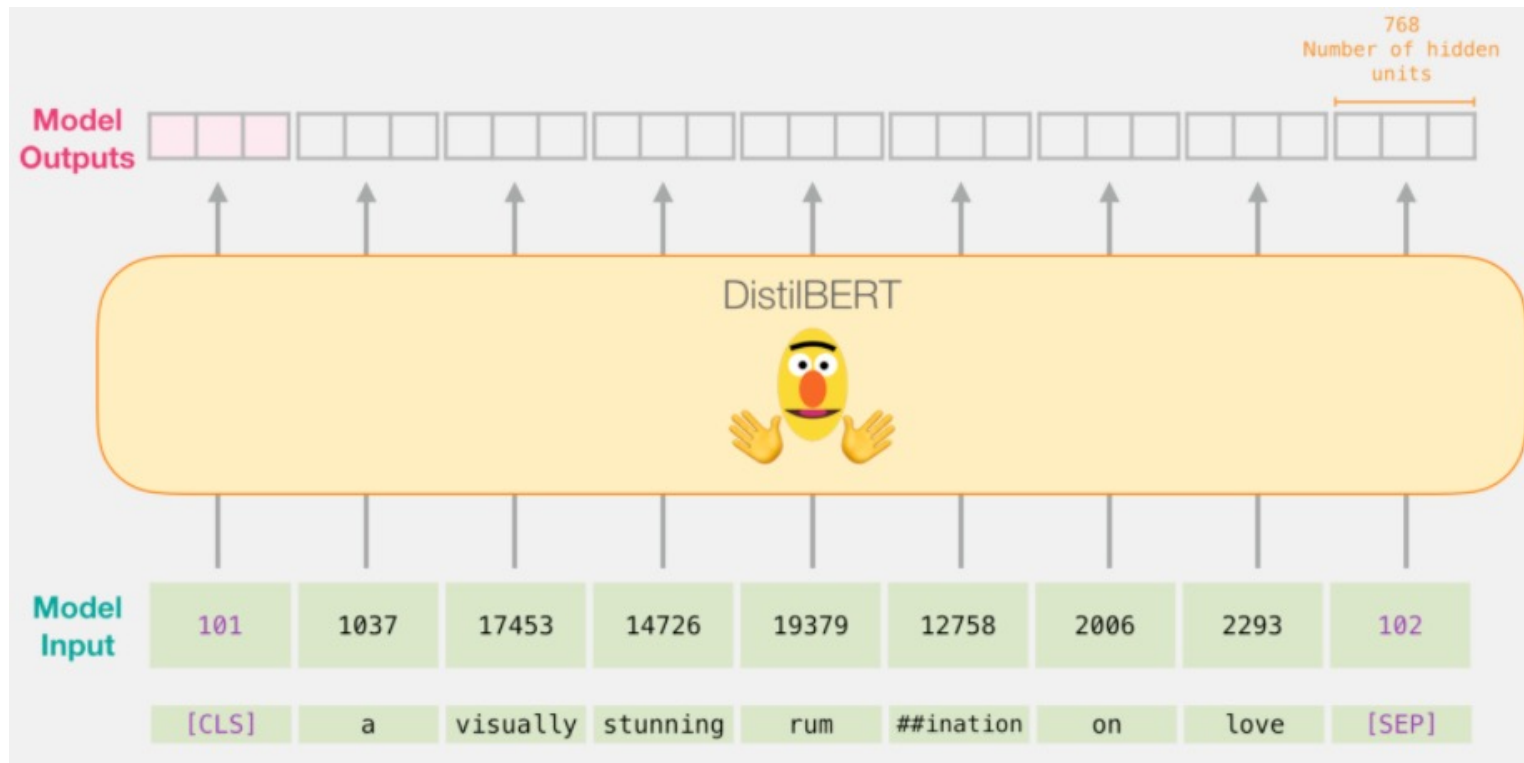
Tokenization



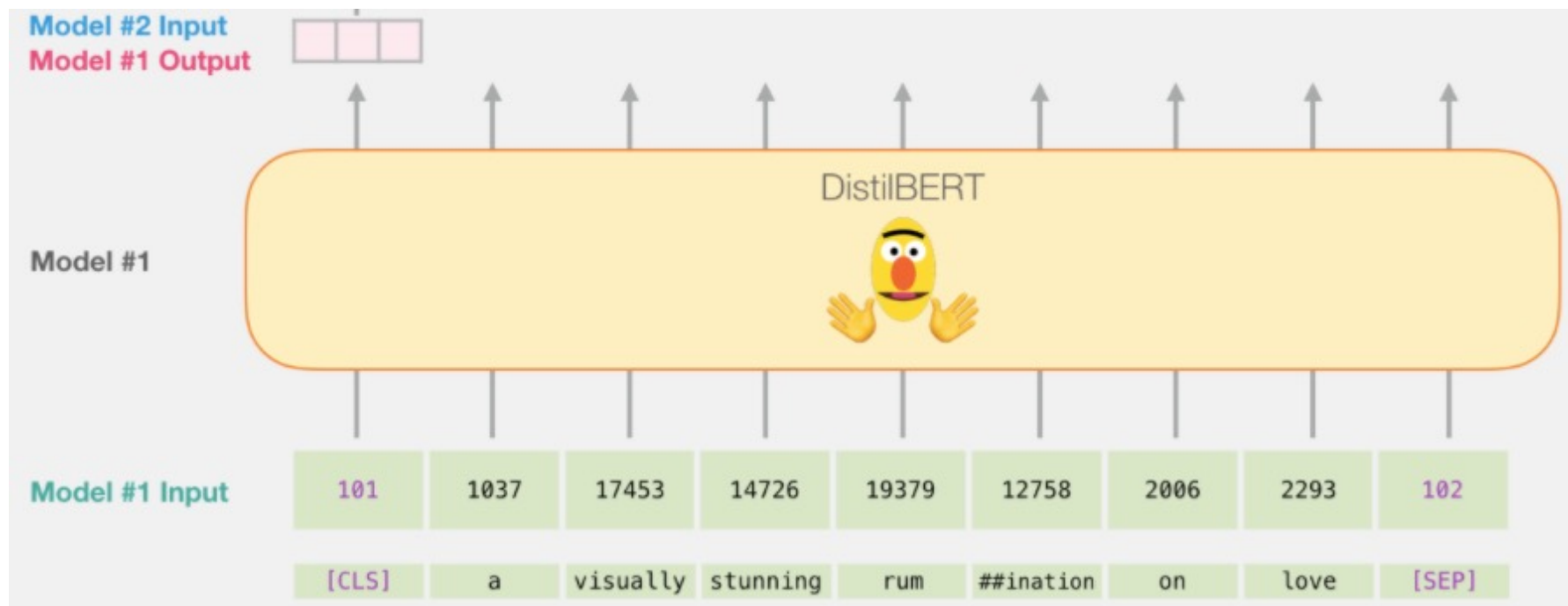
Tokenization



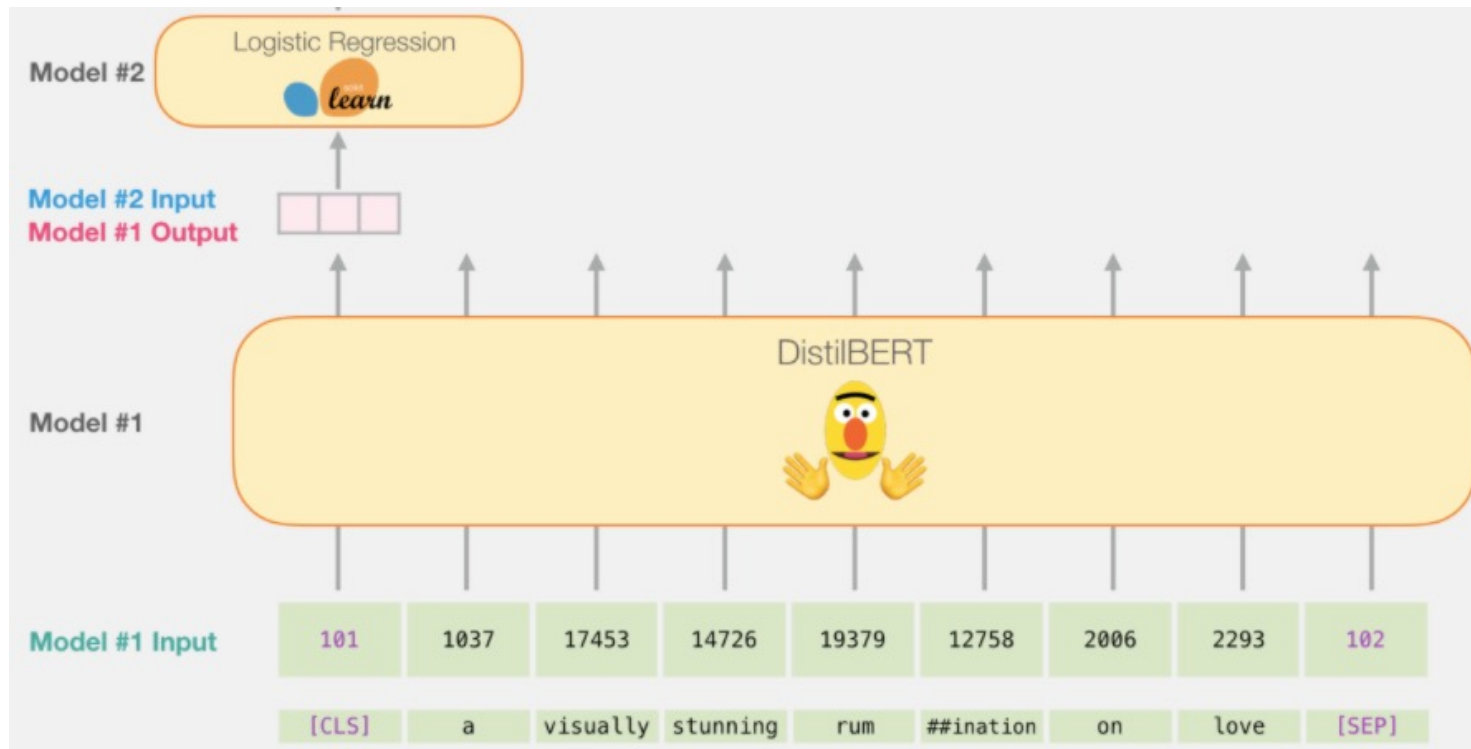
Tokenization



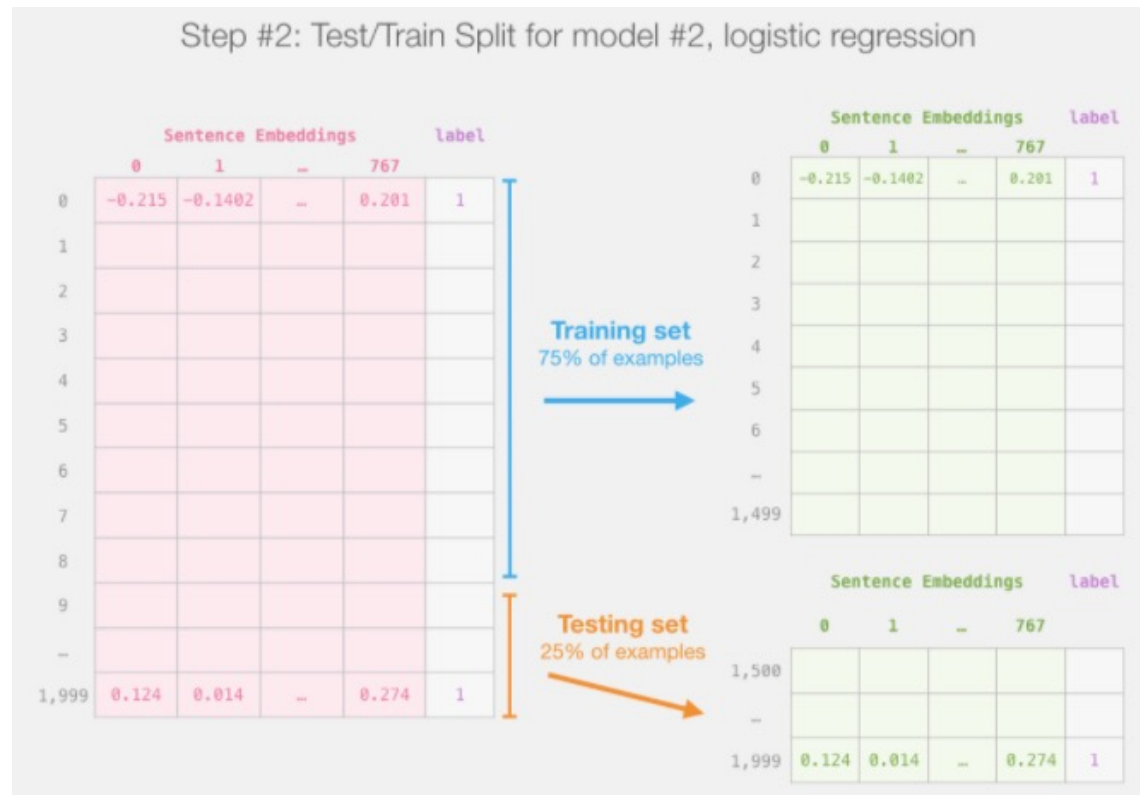
Model 1 : [CLS] Token Embeddings



Model 2: Linear Classifier



Linear Classifier



Linear Classifier

Step #3: Train the logistic regression model using the training set

	Sentence Embeddings				label
	0	1	...	767	
0	-0.215	-0.1402	...	0.201	1
1					
2					
3					
4					
5					
6					
...					
1,499					

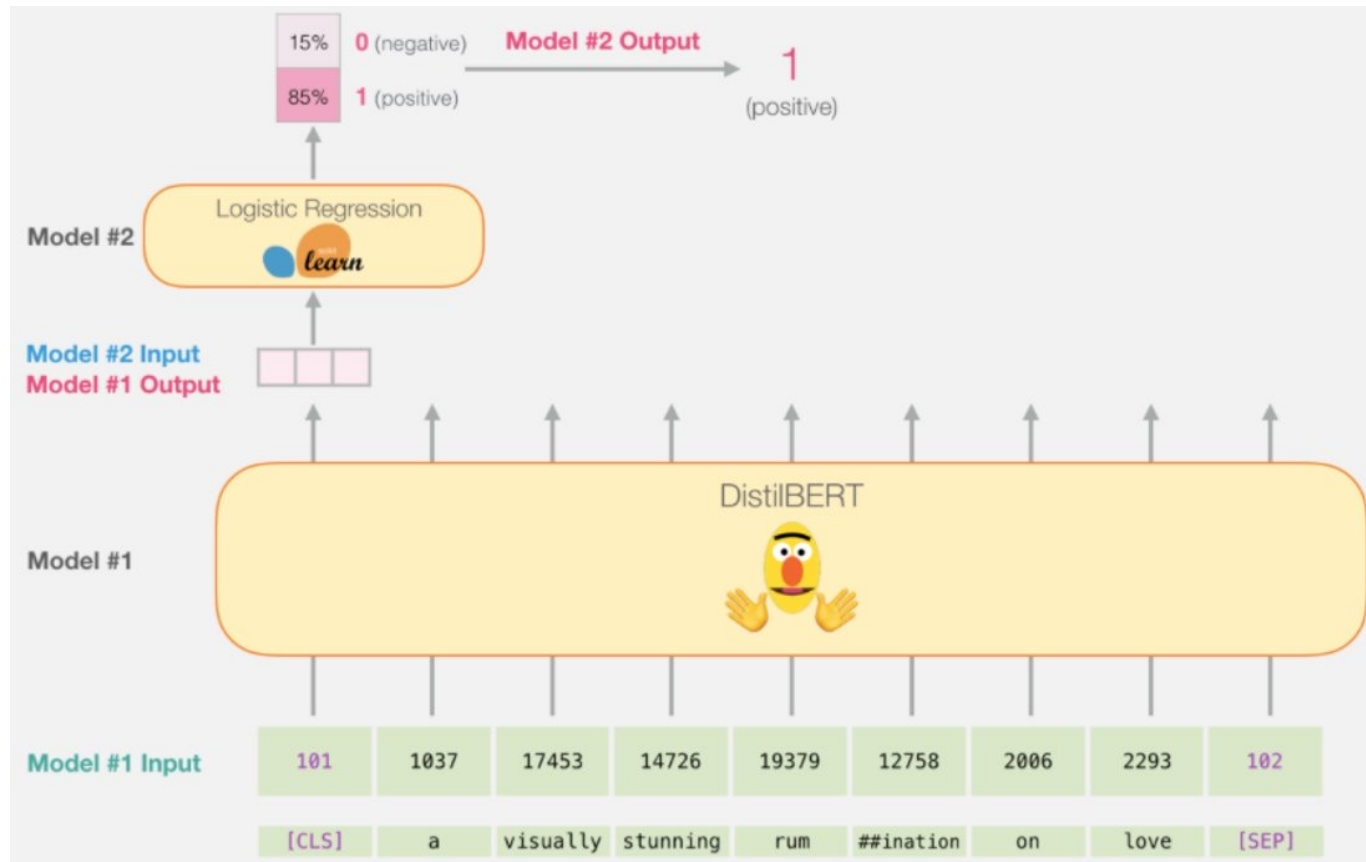
Model
Training



Logistic
Regression



Review Classifier



Finetuning



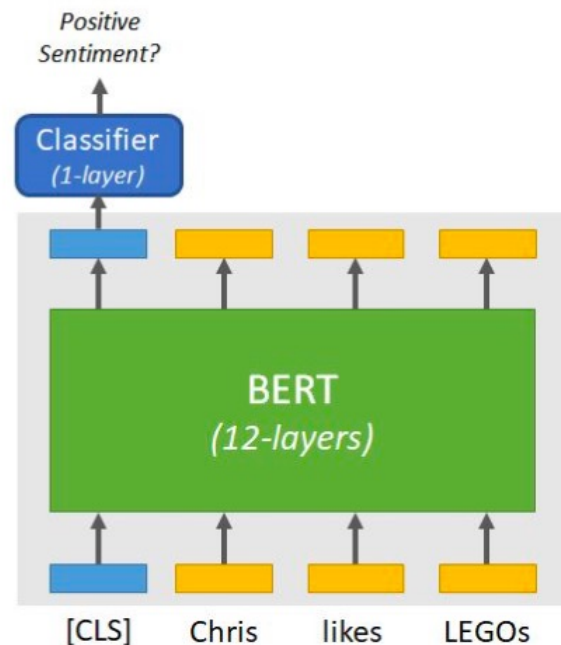
Fine-Tuning

BERT – 12 layer NN

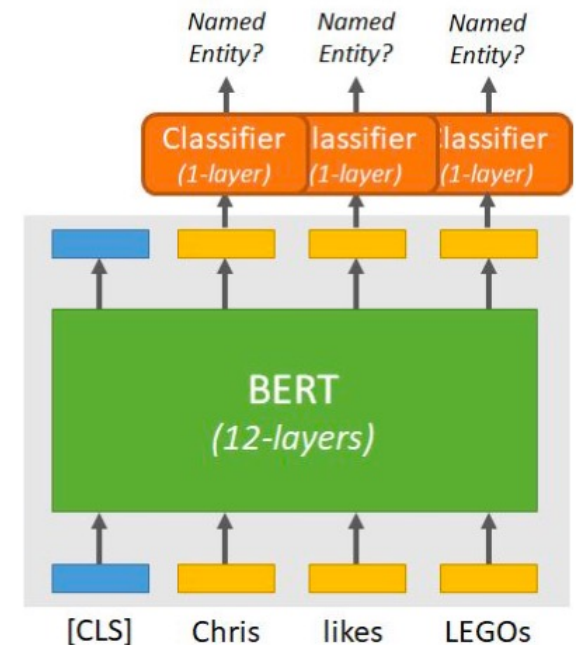
MLM & NSP adds a small, single-layer classifier to get outputs

Magic – You can replace this final layer with your own task-specific model leveraging all BERT's knowledge

Sentiment Analysis



Named Entity Recognition



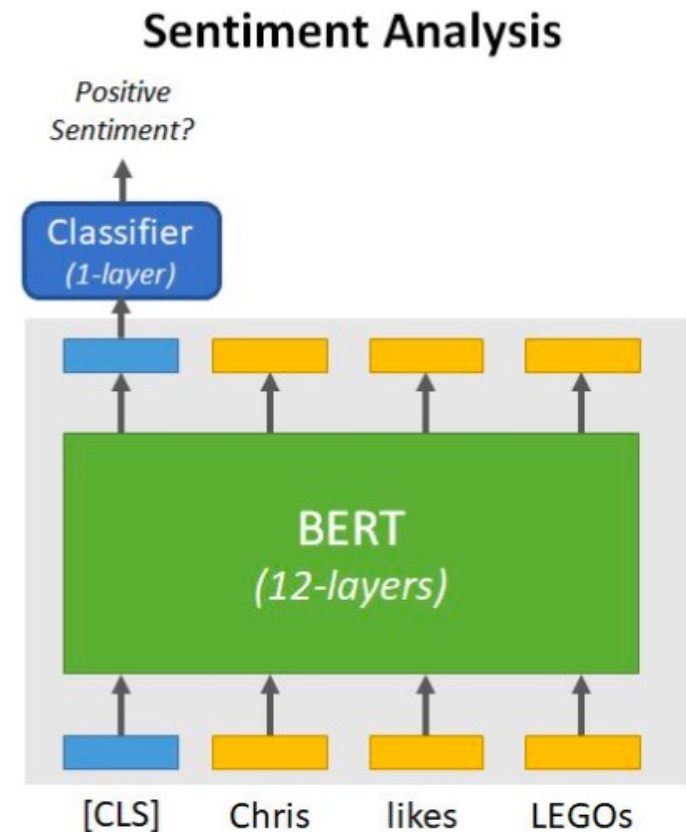
Finetuning Text Classification



Sentiment Analysis

Text Classification Task – Sentiment Analysis

1. Take a pretrained BERT
2. Use only the CLS token embedding output
3. Add an untrained layer of neurons on the end
4. *Train the new, combined model on the custom dataset and task – **Fine tuning***

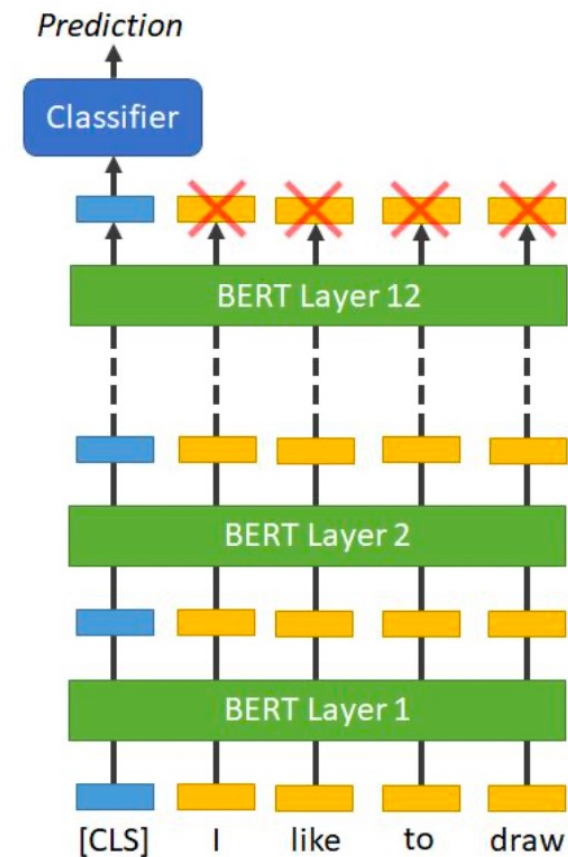


Text Classification

When BERT is fed an input text,

- First token is a **[CLS]** token
- This token is propagated to other layers until the last layer
- Add a single-layer linear classifier only taking this [CLS] BERT embedding as input and providing a classification output
- All other token embeddings are **omitted**

Note: [CLS] token would've been enhanced by the other words (context) in all 12 layers



Text Classification

Youtube Spam Collection Data set

<http://archive.ics.uci.edu/ml/datasets/YouTube+Spam+Collection>

Dataset	YouTube ID	# Spam	# Ham	Total
Psy	9bZkp7q19f0	175	175	350
KatyPerry	CevxZvSJLk8	175	175	350
LMFAO	KQ6zr6kCPj8	236	202	438
Eminem	uelHwf8o7_U	245	203	448
Shakira	pRpeEdMmmQ0	174	196	370



Finetuning NER

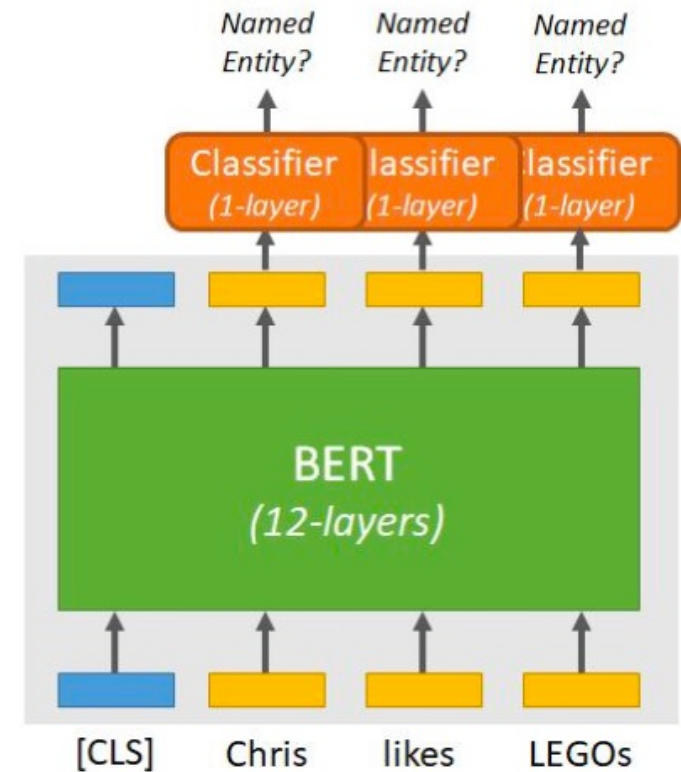


NER

Token Classification Task – NER

1. Take a pretrained BERT
2. Use all output but CLS output
3. Add an untrained layer of neurons on the end for every input word
4. *Train the new, combined model on the custom dataset and task – **Fine tuning***

Named Entity Recognition



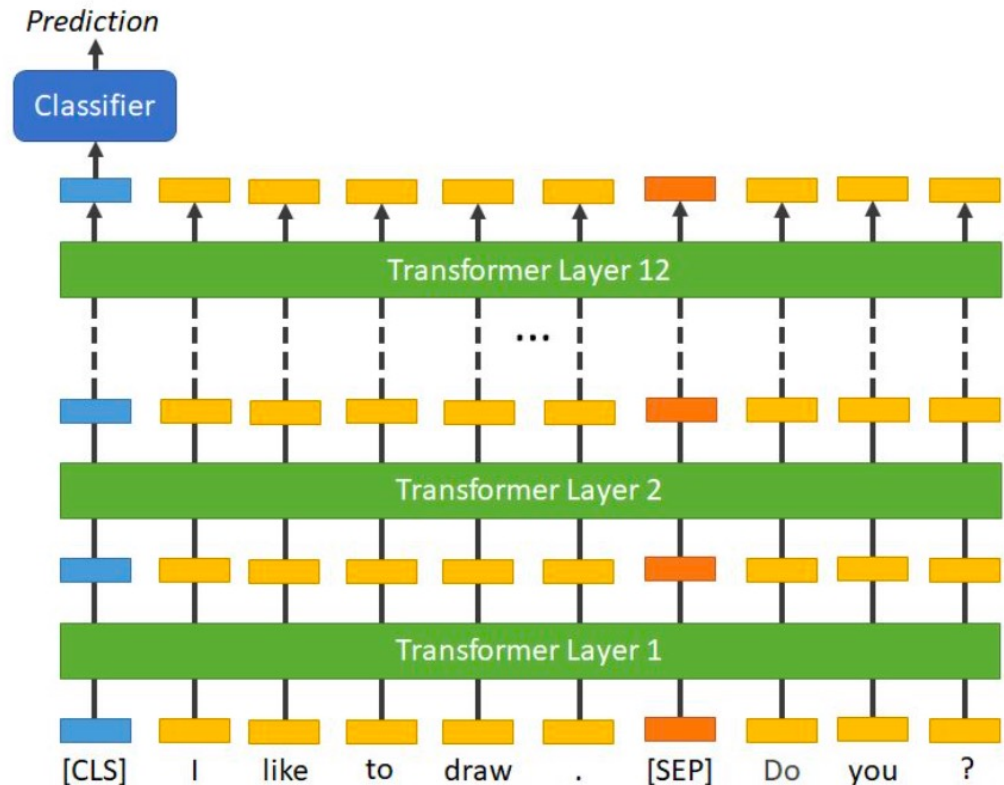
Finetuning QA



Text Pair Classification

When BERT is fed an input text,

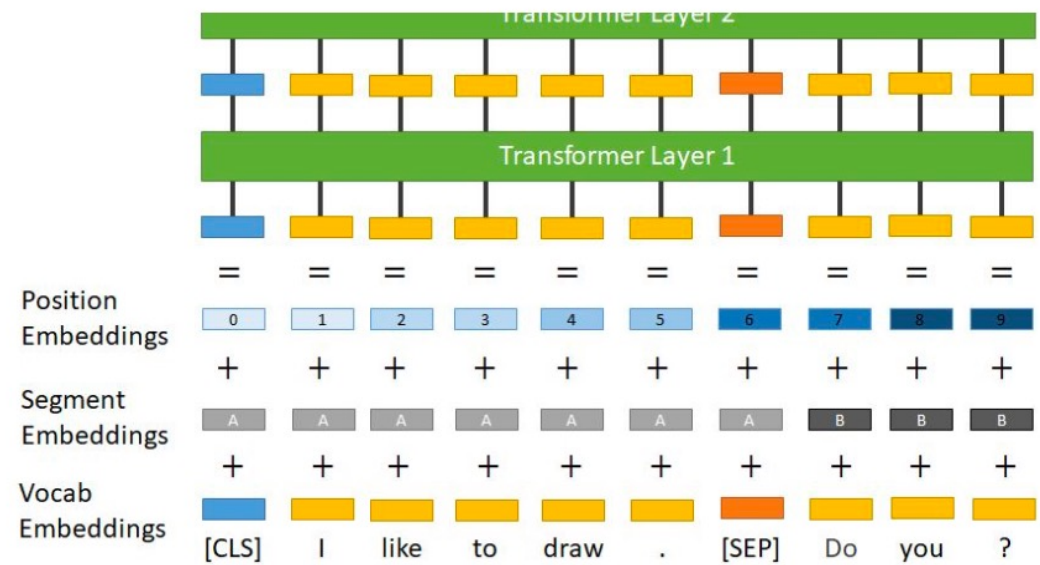
- First token is a **[CLS]** token
- **[SEP]** token is sent as sentence separator
- Add a single-layer linear classifier only taking this [CLS] BERT embedding as input and providing a classification output
- All other token embeddings are **omitted**



Text Pair Classification

When BERT is fed an input text,

- First token is a **[CLS]** token
- **[SEP]** token is sent as sentence separator
- **Segment embeddings** are added for every sentence
- **Positional Embeddings** are added for every word
- Add a single-layer linear classifier only taking this [CLS] BERT embedding as input and providing a classification output
- All other token embeddings are **omitted**



Further Reading

Transformers

<https://huggingface.co/transformers>

BERT

<https://arxiv.org/pdf/1810.04805.pdf>

<https://github.com/google-research/bert>

